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Assignment -1
SEPM

Q1) Explain software models.

Ans: (1) A software process model is an abstract representation of a software development process. It defines the specific flow of activities, actions, and tasks required to build high-quality software. The primary goal is to provide structure to what would otherwise be a chaotic development.

(2) Fundamental process activities
Every model, regardless of its structure, must address these four fundamentals activities:
(1) Software specification
(2) Software Development
(3) Software Validation
(4) Software Evolution

(1) Waterfall Models:
It is a famous and good version of SDLC (System Development Lifecycle) for software engineering. The waterfall model is a linear and sequential model, which means that a development phase cannot begin until the previous phases is completed. We cannot overlap phases in waterfall Model.

Key characteristics:
(1) Each phase must be "signed off" and complete before the next ones begin.
(2) Phases: Requirement Analysis, System Design, Implementation, Integration & Testing,

Deployement & maintenance.

Advantages:
- simple to understand
- easy to manage

Disadvantage:
- very difficult to go back and change requirements
- no working software is produced until late in the lifecycle.

Project planning → Requirement → Analysis → Design → Loading → Testing → Deployement

Incremental model:

(2) ~~Spiral~~ Model: ~~The~~ spiral ~~model~~ is an ~~evolutiona-ry~~ model that each "build" or increment must pass through these specific stages:

(1) Requirement Analysis: The overall system requirements are defined, then specific features are assigned to a particular increment.

(2) Design: The architecture of that specific module is planned to ensure it fits with the existing core.

(3) Development : The actual programming of the module takes place.

(4) Testing: The new increment is tested individually & then integrated with previous versions to check for compatibility.

(5) Deployement: The functional part is delivered to the customer for use & evaluation.

- Advantages:
  (i) Users get a working software in hand quickly.
- Disadvantages:
  (i) High planning - It requires a clear "big picture" before starting.

Requirements → Design & Development → Testing → Implementation
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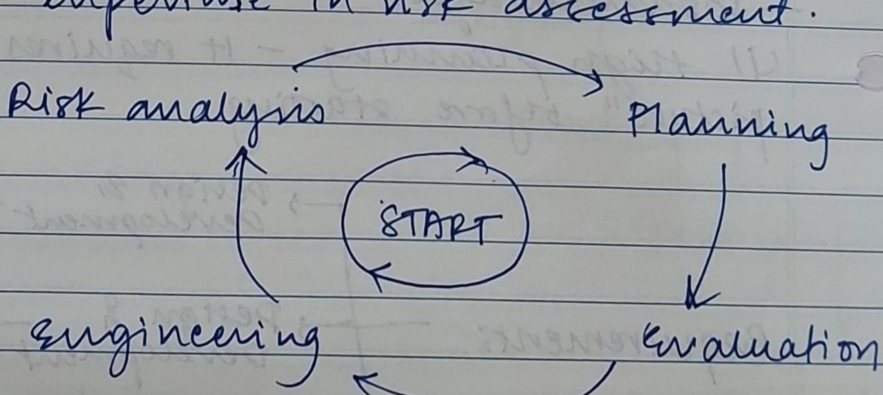
(3) Spiral model: The spiral model is an evolutionary model that combines iterative nature of prototyping with the controlled aspects of the waterfall model.

Key characteristics: It is divided into 4 quad
The starts in the center and moves in loops.
loops involves:
(1) Planning: Determining objectives and constraints
(2) Risk analysis: Identifying and resolving technical and management risks.
(3) Engineering: Developing the next level of the product
(4) Evaluation: customer reviews the progress before the next spiral.

<u>Advantages</u>: strongest model for risk management, suitable for large complex system.

<u>Disadvantage</u>: can be very expensive, requires high expertise in risk assessment.



(4) Concurrent model: The model is viewed as a network of activities. Each activity exists in a specific state at any given time. For example, the "Design" activity might be in the under Development state, while the

"Requirements" activity is in the awaiting respo
nse state.

In this model, "phases" are tested as concurrent activities that communicate with each other:
(i) communication: gathering requirements from the client.
(ii) analysis: Modeling the data and functional requirements.
(iii) Design: creating the technical architecture
(iv) Construction: coding & building the system
(v) validation: continuous testing as components are completed.

- <u>Advantage</u>: significantly reduces the total time to market because tasks overlap.
- <u>Disadvantages</u>: very difficult to manage, requires constant communication bet. tea